Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Practical	Tutorial	Theory	Practical /Oral	Tutorial	Total
2014113	**Operating System**	03	--	--	03	--	--	03

Course Code	Course Name	Examination Scheme						
		Theory Marks				Term Work	Pract. /Oral	Total
		Internal assessment			End Sem. Exam			
		Test1	Test 2	Total				
2014113	**Operating System**	20	20	40	60	--	--	100

Course Objectives:

Sr. No.	Course Objectives
The course aims:	
1	To understand the basic concepts of Operating System, its functions and services.
2	To introduce the concept of a process and its management like transition, scheduling, etc.
3	To understand basic concepts related to Inter-process Communication (IPC) like mutual exclusion, deadlock, etc. and role of an Operating System in IPC.
4	To understand the concepts and implementation of memory management policies and virtual memory.
5	To understand functions of Operating System for storage management and device management.
6	To study the need and fundamentals of special-purpose operating system with the advent of new emerging technologies.

Course Outcomes:

Sr. No.	Course Outcomes
1	Define the basic concepts of Operating System, its operations and services.
2	Explain the process management policies and describe the scheduling of processes by the Operating System.
3	Apply synchronization primitives to address process coordination and demonstrate the occurrence of deadlock conditions.
4	Analyze memory allocation and management functions of Operating System.
5	Evaluate the effectiveness of the services provided by the Operating System for File and I/O Management, considering their impact on overall system performance.
6	Design a framework to compare and optimize the functions of various special-purpose Operating Systems for specific application requirements.

DETAILED SYLLABUS:

Sr. No.	Module	Detailed Content	Hours	CO Mapping

I	Fundamentals of Operating System	**Introduction of Operating Systems:** System Boot, Objectives of Operating System Functions of Operating System, Operating System Structure and Operations, Operating System Services, Multiprogramming, Multitasking, Multithreading, Types of Operating System, Types of System Calls.	**03**	CO1
		Self-learning Topics: Study of various Operating System Architecture like IoT, Android.		
II	Process Management	Basic Concepts of Process: Process State Transition Model, Operations, Process Control Block**,** Context Switching; Introduction to Threads, Types of Threads, Thread Models**,** Basic Concepts of Scheduling, Types of Schedulers, Type of scheduling algorithms: Preemptive and non preemptive (FCFS, SJF, Priority and Round Robin) **Self-learning Topics:** Real-time Scheduling algorithms and applications.	**06**	CO2
III	Process Synchronization and Deadlock Management	Basic Concepts of Inter-process Communication and Synchronization, Race Condition, Critical Section Problem ,Peterson’s Solution, Process Synchronization, Hardware and Semaphores, Producer Consumer Problem. Deadlocks Management: System Model, Deadlock Characterization, Deadlock Prevention, Deadlock Avoidance: Bankers algorithm, Deadlock Detection and Recovery. **Self-learning Topics:** Study a real time case study for Deadlock detection and recovery. Overview of security mechanism in OS.	**10**	CO3
IV	Memory Management	Basic Concepts of Memory Management: Swapping, Memory Allocation strategy, Paging, Structure of Page Table, Segmentation, TLB. Basic Concepts of Virtual Memory, Demand Paging, Copy-on Write, Page Replacement Algorithms, Thrashing. **Self-learning Topics:** Memory Management of IoT, Android Operating System.	**09**	CO4
V	File and IO Management	File Management: Basic Concepts of File System, File Access Methods, Directory Structure, File-System implementation, Allocation Methods, Overview of Mass-Storage Structure, I/O devices, Organization of the I/O Function, Disk Organization, I/O Management and Disk Scheduling: FCFS, SSTF, SCAN, C-SCAN, LOOK, C-LOOK. **Self-learning Topics:** File System for Linux and Windows, Features of I/O facility for different OS.	**07**	CO5

VI	Special-purpose Operating Systems	Open-source and Proprietary Operating System, Fundamentals of Distributed Operating System, Network Operating System, Architecture and functions: Cloud Operating System, Real-Time Operating System, Mobile Operating System. **Self-learning Topics:** Case Study on any one Special-purpose Operating Systems.	**04**	CO6

Text Books:

1. A. Silberschatz, P. Galvin, G. Gagne, Operating System Concepts, 10th ed., Wiley, 2018.
2. W. Stallings, Operating Systems: Internal and Design Principles, 9th ed., Pearson, 2018.
3. A. Tanenbaum, Modern Operating Systems, Pearson, 4th ed., 2015.

Reference Books:

1. Achyut Godbole and Atul Kahate, Operating Systems, McGraw Hill Education, 3rdEdition
2. N. Chauhan, Principles of Operating Systems, 1st ed., Oxford University Press, 2014.
3. A. Tanenbaum and A. Woodhull, Operating System Design and Implementation, 3rd ed., Pearson.
4. R. Arpaci-Dusseau and A. Arpaci-Dusseau, Operating Systems: Three Easy Pieces, CreateSpace Independent Publishing Platform, 1st ed., 2018.

Online References:

1. https://www.nptel.ac.in
2. https://swayam.gov.in
3. https://www.coursera.org/

Assessment:

Internal Assessment (IA) for 40 marks:

IA will consist of Two Compulsory Internal Assessment Tests. Approximately 40% to 50% of syllabus content must be covered in First IA Test and remaining 40% to 50% of syllabus content must be covered in Second IA Test.

End Semester Internal Examination for 40 marks:

Question paper format:

- Question Paper will comprise of a total of **six questions each carrying 20 marksQ.1** will be **compulsory** and should **cover maximum contents of the syllabus**
- **Remaining questions** will be **mixed in nature** (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of **Three questions** needs to be answered.